1. Systematic Encoding ($\mathbf{x} = \mathbf{m}G_1$)

$$\mathbf{m} = \overset{1\times 4\text{ message}}{\begin{pmatrix} m_1 & m_2 & m_3 & m_4 \end{pmatrix}} \times \overset{4\times 5\text{ Generator Matrix } G_1}{\begin{array}{c} \begin{array}{ccccc} & I_4 & & & P \end{array} \\ \left(\begin{array}{cccc:c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right) \end{array}} = \overset{1\times 5\text{ Codeword }\mathbf{x}}{\begin{pmatrix} m_1 & m_2 & m_3 & m_4 & \Sigma m_i \end{pmatrix}}$$

2. The Null Space Condition ($H_1\mathbf{x}^T = \mathbf{0}$)

$$H_1 = \overset{1\times 5\text{ Parity-Check Matrix}}{\begin{array}{c} \begin{array}{ccccc} & -P^T & & & I_1 \end{array} \\ \left(\begin{array}{cccc:c} 2 & 2 & 2 & 2 & 1 \end{array}\right) \end{array}} \times \overset{\mathbf{x}^T}{\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix}} = \mathbf{0} \pmod 3$$

3. Erasure Recovery (Dot Product)

$$H_1 = \begin{pmatrix} 2 & \boxed{\mathbf{2}} & 2 & 2 & 1 \end{pmatrix} \times \overset{\text{Received } \mathbf{y}^T}{\begin{pmatrix} x_1 \\ \boxed{?} \\ x_3 \\ x_4 \\ x_5 \end{pmatrix}} = \mathbf{0} \pmod 3$$

Isolates unknown

The Linear Algebra Perspective:

The dot product expands to:
$2x_1 + 2(?) + 2x_3 + 2x_4 + x_5 \equiv 0 \pmod 3$

Because the column in H_1 corresponding to the erasure is **non-zero** (it is a 2), the unknown variable is not multiplied by zero.

It forms a solvable linear equation of rank 1, guaranteeing a unique solution in $\mathbb{F}_3$.